

Welcome to the Supply Chain Disruption Game!

Practice Session Code: knxm

In this game, your team will play the role of a sourcing manager who is in charge of purchasing a (generic) product. All members in your team make decisions as one single player in the game. There are two potential suppliers for the product: an Overseas supplier and a Local supplier. The computer plays the role of the suppliers. You will play the game for multiple rounds, with each round representing one week. You must determine which supplier(s) to source the product from and how much to source from the supplier(s) each week.

At the beginning of the game (week 0), you will make a one-time setup decision of which supplier(s) to source from. The cost to set up sourcing with a supplier is \$3,000. You must select at least one supplier to source from and you may select both suppliers. During the game, you can only order from the supplier(s) that you have set up sourcing with; you may place orders throughout all weeks of the game (except from the Overseas supplier when a supply disruption occurs; see below for more details). Once made, the supplier selection is fixed for the entire duration of the game.

The Overseas supplier has unlimited capacity but is subject to potential supply disruptions. In each week, there is a 10% chance that disruption may occur at the Overseas supplier, and if occurred, the disruption would last for 3 weeks. During the weeks of disruption, you cannot order from the Overseas supplier, although orders that you have previously placed (before the disruption) will continue to arrive. The Local supplier has a capacity of 500 units per week and is not subject to supply disruptions. The Overseas supplier's lead time is 3 weeks, while the Local supplier's lead time is 1 week.

It is less expensive to source from the Overseas supplier. The per-unit purchasing cost is \$5.00 for the Overseas supplier and \$6.50 for the Local supplier. Purchasing costs are paid to the supplier(s) when you place the respective orders.

At the start of the game, you will be provided with an on-hand inventory of 2,000 units. In each week, demand is fulfilled automatically by the computer. Using your on-hand inventory, the computer attempts to fulfill all or as much of the current-week demand plus any backorders from past weeks. Unfulfilled demand is backordered and incurs a per-unit, per-week backorder cost of \$2.50. Leftover on-hand inventory incurs a per-unit, per-week holding cost of \$1.00.

The costs you will incur in the game are the setup cost for the supplier(s), purchasing costs, inventory holding costs, and backorder costs. **Your objective is to minimize your total cost for the entire duration of the game. The player (team) who incurs the lowest total cost wins.**

Demand information

Customer demand in each week is uncertain and follows a normal distribution with a mean of 600 units and a standard deviation of 250. Each week, demand will be **independently** drawn from this normal distribution. So, whether demand is small or large this week is not affected by demand in past weeks and also has no impact on demand in future weeks.

During the weeks of a disruption at the Overseas supplier, a spike in demand would occur in which the mean of the weekly demand is scaled by a multiplier of 2. The standard deviation of the weekly demand does not change during the demand spike. Once the disruption ends, the weekly demand will fall back to the original normal distribution specified above.

How are the costs calculated?

Your cost in week i is calculated as follows

Week i cost =

$$\begin{aligned} & \$5.00 \times \text{Order quantity accepted by Overseas supplier (if set up) in week } i \\ & + \$6.50 \times \text{Order quantity accepted by Local supplier (if set up) in week } i \\ & + \max(0, \text{Week } i \text{ Demand} + \text{Backorders} - \text{On-hand Inventory at the start of week } i) \times \$2.50 \\ & + \max(0, \text{On-hand Inventory at the start of week } i - \text{Week } i \text{ Demand} - \text{Backorders}) \times \$1.00 \end{aligned}$$

The first two terms represent the purchasing costs paid to the supplier(s), the third term captures the backorder cost, and the fourth term captures the inventory holding cost.

Sequence of events

The sequence of events for week i are as follows:

- 1) At the start of the week, you will receive any inventory you ordered from the Overseas supplier 3 weeks ago and from the Local supplier 1 week ago.
- 2) You will then decide how much to order from the supplier(s). If a disruption occurs (or is still in progress), then you cannot order from the Overseas supplier.
- 3) Demand for the current week is realized. If inventory is available, then the computer automatically fulfills the demand plus any backorders from past weeks up to the total on-hand inventory.
- 4) Your cost for the week is calculated as follows:

Week i cost =

$$\begin{aligned} & \$5.00 \times \text{Order quantity accepted by Overseas supplier (if set up) in week } i \\ & + \$6.50 \times \text{Order quantity accepted by Local supplier (if set up) in week } i \\ & + \max(0, \text{Week } i \text{ Demand} + \text{Backorders} - \text{On-hand Inventory at the start of week } i) \times \$2.50 \\ & + \max(0, \text{On-hand Inventory at the start of week } i - \text{Week } i \text{ Demand} - \text{Backorders}) \times \$1.00 \end{aligned}$$

Frequently Asked Questions

Q: *Why do I still have backorders when it shows me as having enough inventory on-hand to fulfill my backorders?*

A: Every week, the fulfillment of demand and backorders from past weeks occurs at the end of the week, i.e., at Step 3) above. Because of this, you may see that you have both on-hand inventory and backorders on the screen when making ordering decisions at the start of the week.

Q: *Example: If I place an order in week 3 from the Overseas supplier and the lead time is 4 weeks, when will this inventory be available for me to use?*

A: You will receive the units at the start of week 7.

Q: *When a disruption occurs at the Overseas supplier, will I still receive the orders I placed in previous weeks (but have not yet arrived)?*

A: Yes, any orders placed with the Overseas supplier before the disruption occurs will be received.